

Building Agentic AutoML

A progressive educational project for building an Agentic AutoML system step by step

Project Report - Final Roadmap Edition

OBSERVE

DECIDE

ACT

EVALUATE

REMEMBER

RETURN

Final project report - August 2026

1. Executive Summary

Building Agentic AutoML is a progressive educational project that develops an Agentic AutoML system for tabular supervised learning across eight independently executable notebook versions. The project starts from a minimal baseline and ends with a Senior Agent that plans experiments adaptively within a fixed budget instead of evaluating the complete candidate space exhaustively.

The final system detects binary classification, multiclass classification and regression; inspects dataset structure and data quality; compares preprocessing and feature engineering strategies; evaluates LightGBM, XGBoost and CatBoost; compares model-specific hyperparameter configurations; selects task-aware metrics and validation; and preserves a transparent decision history.

Version 8 completes the original roadmap. On Adult Income, the Senior Agent recovers the same best observed configuration found by the exhaustive Version 7 search while executing 10 experiments out of 24 available candidates.

2. Project Goal and Scope

- Educational: make the AutoML decision process visible and understandable directly from the notebooks.
- Progressive: introduce one new capability per version while preserving previous components as much as possible.
- Reproducible: use explicit random seeds, task-aware cross-validation and fold-isolated preprocessing and feature engineering.
- Traceable: store experiment configuration, scores, variability, status, duration and the reason behind adaptive decisions.
- Practical: work on realistic tabular classification and regression datasets without hiding the modeling logic behind a large AutoML framework.

Scope choices are intentional. The project is not a production AutoML platform and does not attempt to replace mature optimization libraries. Its purpose is to expose how an agent can progressively acquire AutoML capabilities and then use them under a simple planning policy.

3. Educational Method

The project follows one central rule: one version, one concept, one architectural improvement. Every version is complete enough to run end to end and remains available as an independent learning resource. New abstractions are introduced only when they make a concrete new capability visible.

- Notebooks are readable from top to bottom and keep functions small and explicit.
- The main agent function coordinates components instead of hiding their logic.
- Outputs expose raw experiment state and decision history, not only polished summaries.
- The Adult Income benchmark is reused across versions to isolate the effect of each architectural change.
- Additional datasets are introduced when the project needs to demonstrate multiclass or regression generalization.

4. Final Agent Architecture

The final Version 8 architecture separates the experiment space from the planning policy. The same 24 preprocessing-feature engineering-model-hyperparameter candidates introduced in Version 7 remain available, but the Senior Agent selects only a budgeted subset to execute.

Stage	Responsibility
Observe	Detect task and feature types; inspect shape, target, missing values, dtypes, cardinality and duplicates.
Decide	Use the Senior Agent policy to choose the next experiment from the remaining candidates.
Act	Execute the selected preprocessing, feature engineering, model and hyperparameter configuration.
Evaluate	Cross-validate with the selected metric and record fold scores, mean, standard deviation and failures.
Remember	Append experiment metadata, duration, status, decision reason and best-score trajectory.
Return	Report the best observed configuration together with the complete experiment and decision history.

5. Version Roadmap

Version	Name	What changes	Main lesson
V01	Baseline Agent	Task detection, feature detection, LightGBM baseline and structured state.	Minimal end-to-end AutoML agent
V02	Data Inspector	Dataset shape, target, dtypes, missing values and cardinality.	Observe data before acting
V03	Preprocessing Agent	Compare native handling and simple imputation.	Data preparation by evidence
V04	Model Selection	Compare LightGBM, XGBoost and CatBoost.	Model choice by validation
V05	Smart Validation	Use StratifiedKFold or KFold and store mean/std across folds.	Reliable task-aware evaluation
V06	Feature Engineering	Compare original features with numerical interactions.	Controlled feature-space search
V07	Hyperparameter Optimization	Compare baseline and one alternative configuration per model.	Treat parameters as an AutoML choice
V08	Senior Agent	Plan which experiments to execute within a fixed budget.	Adaptive and transparent experiment planning

The roadmap is intentionally compact. Version 8 does not replace the tools introduced earlier; it changes how the available experiment space is explored. This preserves the One Version, One Concept principle through the final stage.

6. Dataset Strategy

The project uses a compact Kaggle dataset collection designed to increase difficulty progressively. The same agent interface receives a CSV path and a target column; task type and feature types are detected by the code.

File	Task	Type	Purpose
simple_binary.csv	Binary classification	Synthetic	Early controlled benchmark with mixed features and missing values.
simple_regression.csv	Regression	Synthetic	Controlled regression benchmark for early task generalization.
adult_income.csv	Binary classification	Real-world	48,842 rows; mixed types, missing values and class imbalance; main development benchmark.
wine.csv	Multiclass classification	Real-world	178 rows, 13 numerical features and 3 classes; validates multiclass support.
california_housing.csv	Regression	Real-world / mixed	20,640 rows; numerical + ocean_proximity; missing total_bedrooms; tests mixed regression.
california_housing_numeric.csv	Regression	Real-world / numeric	20,640 rows; eight numerical predictors with complete numeric observations.

The final Version 8 generalization tests intentionally cover four distinct conditions: binary classification with mixed data and missing values, multiclass classification, mixed-type regression with missing values, and numeric-only regression without missing values.

7. Final System Capability

- Detect binary classification, multiclass classification and regression.
- Identify numerical and categorical features from dataframe dtypes.
- Inspect shape, duplicates, target distribution/summary, missing counts and percentages, dtypes and cardinality.
- Compare native missing-value handling with simple training-fold imputation.
- Compare no feature engineering with pairwise numerical interaction features inside each fold.
- Build LightGBM, XGBoost and CatBoost classifiers/regressors with task-appropriate objectives.
- Compare baseline and alternative model-specific hyperparameter configurations.
- Use ROC AUC, ROC AUC OvR Macro or RMSE according to the detected task.
- Use StratifiedKFold for classification and KFold for regression, adapting classification folds when necessary.
- Track fold scores, mean, standard deviation, duration, status, errors and decision reasons.
- Plan a budgeted sequence of experiments and return the best observed configuration with complete decision history.

8. Version 8 Planning Policy

Stage	Rule
1	Establish native + none + default baselines for LightGBM, XGBoost and CatBoost.
2	Try the alternative hyperparameter configuration of the current best model.
3	When missing values exist, compare the alternative preprocessing while holding the rest of the best configuration fixed.
4	Test numerical interactions around the current best configuration.
5	Use the remaining budget to explore untested candidates deterministically.

The policy is deliberately simple and rule-based. Its educational purpose is to make adaptive experiment selection observable before introducing more advanced optimization or planning mechanisms.

9. Final Version 8 Benchmark Results

Dataset	Task	Metric	Best model	Preproc.	Feature eng.	Score	Std	Sec
Adult Income	binary	ROC AUC	CatBoost	native	none	0.930631	0.002246	812.45
Wine	multiclass	AUC OvR Macro	CatBoost	native	none	0.998847	0.001814	90.39
California Housing	regression	RMSE	CatBoost	impute	interactions	45482.05	1289.98	229.36
California Numeric	regression	RMSE	CatBoost	native	interactions	45703.19	1375.85	86.30

Every benchmark uses a maximum budget of 10 experiments out of 24 available candidates. With 5-fold cross-validation this corresponds to at most 50 model fits per dataset instead of the 120 fits required by the Version 7 exhaustive search.

On Adult Income, Version 8 recovers the same best observed configuration as Version 7: CatBoost + native + none + baseline. The result demonstrates useful search reduction without claiming that the deterministic policy guarantees the global optimum.

10. Limitations and Project Status

- The planning policy is intentionally simple, deterministic and rule-based.
- The budget is an experiment count; execution time is recorded but does not guide choices.
- No Bayesian optimization, reinforcement learning, LLM planning or dedicated early stopping is used.
- Validation is not group-aware or time-series-aware, and no nested cross-validation or final independent holdout is used.
- Task detection remains heuristic for discrete numerical targets.
- Feature engineering is limited to pairwise numerical interactions and only three boosting model families are available.
- The final state does not store a persistent fitted pipeline.

These are intentional scope choices rather than unfinished roadmap items.

Version 8 completes the original roadmap. The project can remain finished in this form, while documentation, compatibility fixes and small maintenance updates may still be made when useful. The repository is intentionally not declared permanently frozen: if a future concept is worth studying with the same one version, one concept, one architectural improvement philosophy, additional versions may be created.

There is currently no required Version 9. Any future continuation would be an optional extension of an already completed project.

Project Infographic

Building Agentic AutoML - Project Roadmap (Versions 1 to 8)

One version, one concept, one architectural improvement.

1 Baseline Agent Baseline end-to-end workflow	2 Data Inspector Observe dataset structure	3 Preprocessing Agent Compare data preparation	4 Model Selection Compare model families	5 Smart Validation Task-aware cross-validation	6 Feature Engineering Compare feature spaces	7 Hyperparameter Optimization Compare model configurations	8 Senior Agent Plan experiments within a budget
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Final Version 8 Architecture

Dataset + Target	Task / Feature Detection	Data Inspection
Candidate Space (24)	Senior Agent Planner	Selected Experiment
Cross-Validation	Decision History / State	Best Observed Configuration

Version 8 completes the original roadmap. The project is complete. There is currently no required Version 9; any future version would be an optional extension of an already completed project.